

Madhav Yadav

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Education

B.Tech in Computer Science — AKTU

📅 Sept 2023 - present

Experience

Agnirath Aerolabs — ML & Software Engineer Intern

📅 Oct 2025 - Jan 2026

Python Django Computer Vision Transformers Next.js UAVs

- Architected and deployed a real-time Ground Control Station (GCS) for UAV monitoring and autonomous mission management, integrating telemetry pipelines with sub-second latency.
- Built high-performance FastAPI backend services in Python to ingest, process, and relay drone flight data and command payloads across embedded UAV systems.
- Developed responsive Next.js operator dashboards visualizing real-time drone telemetry — position, altitude, battery, and mission status — reducing operator response time significantly.

Featured Project

Open-Papertrade

Django Next.js Typescript REST APIs

An AI-powered paper-trading platform to practice trading strategies with real market data and zero financial risk. github.com/Open-Papertrade/Open-Papertrade

- Engineered a full-stack paper trading platform simulating real-world market conditions, enabling users to test algorithmic trading strategies risk-free before live deployment.
- Built robust REST API backend with Django for order management, portfolio tracking, and real-time market data ingestion, supporting concurrent user sessions.
- Designed the system with extensibility in mind — architecture supports future integration of ML-based trading signal models and backtesting engines.

🔗 see all of my projects on my portfolio: [Madhav's Portfolio](#)

Projects

Github MCP Server

Python FastMCP GitHub API LLM Tools

A lightweight Model Context Protocol (MCP) server built with FastMCP to expose and automate GitHub operations using LLM-compatible tools. Some of its current features include: [Github Link](#)

- Retrieve a GitHub user's bio.
- Recursively fetch all files and directories from a GitHub repo starting at a path.
- Fetches the content of a file from a GitHub repository.
- Creates a new branch and then a pull request on GitHub.
- Creates an issue on a GitHub repository.

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Steganographic Encoder/Decoder

Python Django Pillow OpenCV

A Django-based web application that lets users hide and retrieve secret messages within images, providing a practical introduction to the concept of steganography. [Github Link](#)

- Encode hidden messages into images.
- Decode secret messages from images.
- User-friendly web interface built with Django.

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Skills & Tools

Core LLM / Applied AI

- Large Language Models (LLMs)
- Prompt Engineering
- Context Engineering
- Retrieval-Augmented Generation (RAG)
- Agentic RAG
- Hybrid Search
- Reranking
- Vector Search / Semantic Search
- Embeddings
- AI Agents
- Agentic Workflows
- Tool Use / Function Calling
- Model Context Protocol (MCP)
- Structured Outputs
- Multimodal AI
- Fine-tuning
- LoRA / QLoRA
- RLHF / DPO
- Model Distillation

Evaluation & Quality

- LLM Evaluation (Evals)
- LLM-as-Judge
- Offline & Online Evaluation
- Regression Testing for LLMs
- Guardrails
- Hallucination Mitigation
- Prompt Injection Defense
- Responsible / Safe AI
- OWASP LLM Top 10

AI Infra / Platform

- LLM Inference Optimization
- Model Serving
- vLLM
- TGI
- SGLang
- KV Cache
- Continuous Batching
- Quantization (GPTQ/AWQ/GGUF, INT4/INT8/FP8)
- Speculative Decoding
- GPU Optimization
- Vector Databases (pgvector, Qdrant, Weaviate, Milvus)
- ANN Indexing (HNSW/IVF)

LLMOps / MLOps

- LLMOps
- MLOps
- Observability / Tracing (Langfuse, Phoenix, Braintrust)
- Prompt Versioning
- Semantic Caching
- Prompt Caching
- Model Routing / Gateway (LiteLLM)
- Cost & Latency Optimization
- Model Monitoring
- CI/CD for ML

Frameworks & Libraries

- LangChain
- LangGraph
- LlamaIndex
- DSPy
- Hugging Face Transformers
- PyTorch
- pydantic
- FastAPI
- Anthropic / OpenAI SDKs
- sentence-transformers

Languages & Engineering

- Python
- TypeScript / JavaScript
- SQL
- async / concurrent programming
- REST APIs
- Git
- System Design
- Distributed Systems

Cloud & Deployment

- AWS / GCP / Azure
- Docker
- Kubernetes
- Serverless
- GPU Compute
- Postgres
- Redis
- CI/CD

ML Foundations

- Machine Learning
 - Deep Learning
 - Neural Networks
 - Transformers
 - NLP
 - Model Evaluation Metrics
 - Regularization (L1/L2, weight decay)
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